

Machine Learning in Hollow Core Fiber Design

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Abstract We generate numerous high-performance designs using neural networks predicting confinement loss in nested anti-resonant nodeless fibers. These designs provide meaningful statistics to identify a high-fidelity figure of merit, aiding future complex designs with multiple metrics. ©2026 The Author(s)

Introduction

Hollow core fibers (HCFs), specifically nested anti-resonant nodeless fibers (NANFs) [2], offer ultra-low latency, reduced chromatic dispersion, and record-low attenuation [3]. However, the geometric complexity of non-touching nested capillaries introduces rigorous design and fabrication bottlenecks. Recent studies of hybrid HCF-single mode fiber (SMF) systems demonstrate that leveraging near-zero nonlinearity can double transmission capacity even with higher-loss HCF [4]. To enable HCF adoption, we need tools to identify designs robust to fabrication tolerances while balancing competing performance targets: single mode operation, low bending and fundamental mode confinement loss, etc.

We navigate the high-dimensional NANF design space with a neural network (NN) digital twin in place of conventional, computationally expensive finite element method (FEM) simulations. Our NN, trained on 4-node NANF data (geometry in Fig. 1), predicts confinement loss with high fidelity and, crucially, can extrapolate into loss regimes significantly lower than those in the training set [1].

The computational efficiency of the NN enables the exploration of an immense number of designs. From this we extract a large set of high-performance designs whose confinement loss (CL) can be verified via FEM. By exploiting the statistical depth of this set, we uncover common geometric hallmarks that define a new figure of merit (FOM) for NANF. This data-driven framework offers a scalable approach that becomes increasingly essential for navigating the high-dimensional design spaces of complex structures such as double-nested antiresonant nodeless fiber (DNANF), 5-node architectures, etc.

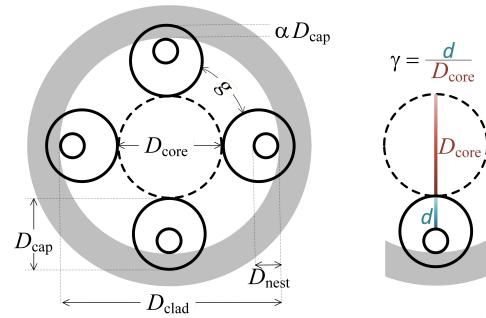


Fig. 1: NANF geometry for proof-of-concept study

NN to Find High-performance Designs

The underlying NN model uses a two-stage architecture: a classifier identifies single-mode behavior (higher-order mode (HOM) suppression ≥ 50 dB), a regressor predicts the logarithm of the CL to manage high dynamic range. This strategy allowed the NN to successfully identify optimized designs with confirmed CL as low as 0.25 dB/km despite being trained on designs with CL above 1 dB/km; see methodology in Fig. 2 [1].

We categorize our four-node NANF designs into three distinct sets whose CL at 1400 nm was confirmed via FEM simulation. Designs in the primary data set are a subset of those used to train the neural network; we retain only 6,927 designs with CL capped at 100 dB/km as those with higher loss are less relevant to our objectives. We also use a set of 234 designs with CL ranging from 0.276 to 1 dB/km found via initial simulations, but excluded from NN training; they provide designs with intermediate performance. Finally, by exploring an immense search space we obtain a set of 362 high-performance designs identified via NN with CL ranging from 0.250 to 0.334 dB/km.

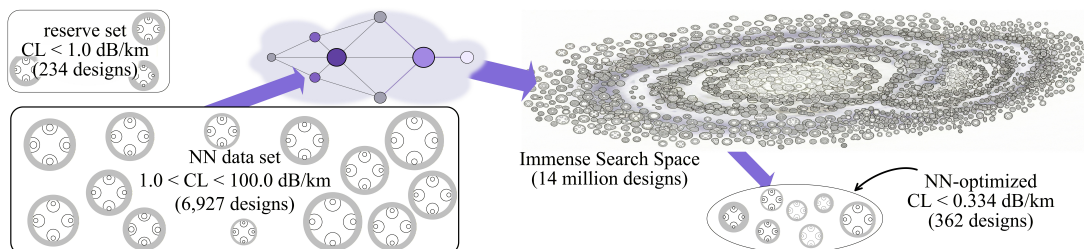


Fig. 2: Methodology for exploiting NN predictions of CL [1].

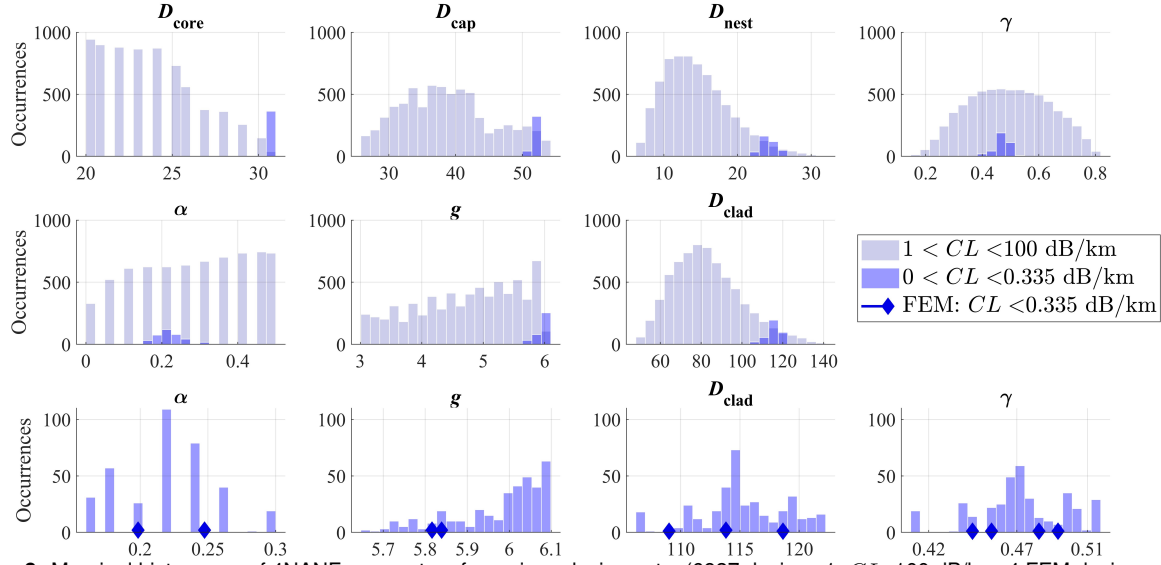


Fig. 3: Marginal histograms of 4NANF parameters for various design sets. (6927 designs $1 < CL < 100$ dB/km, 4 FEM designs and 362 NN designs with $CL < 0.335$ dB/km.

Diversity of NN High-performance Designs

The geometry of a nested antiresonant nodeless fiber (NANF) can be described by a collection of physical dimensions defined in Fig. 1. One straight-forward description is a 4-dimensional vector $[D_{core}, D_{cap}, \alpha, D_{nest}]$, with D_{clad} and g treated as dependent variables. We introduce

$$\gamma = [(1 - \alpha)D_{cap} - D_{nest}] / D_{core}, \quad (1)$$

an intuitive parameter conceptually related to the D_{cap}/D_{core} ratio for standard antiresonant nodeless fiber (ANF) designs, but adapted to account for the nested capillary geometry.

In the first two rows of Fig. 3 we produce the marginal distributions of all 4NANF parameters for the designs we examined. The lighter histogram includes designs with $1 < CL < 100$ dB/km; the darker bars are for the high-performance designs. To ensure manufacturability and favor good performance, we only retained designs for our examination if they adhered to basic design rules [1], e.g., maintaining a gap $3 < g < 6 \mu\text{m}$ to balance resonance and leakage losses. These rules inevitably introduce correlations, even among independent parameters. Hence, although we initiate our search with a uniform distribution for parameters, the design rules deform the distributions in the $1 < CL < 100$ dB/km set.

We can see in the first two rows of Fig. 3 that the high-performance designs have very distinct distributions from those imposed by basic design rules. Row three of Fig. 3 highlights the importance of using NN techniques to uncover this behavior. During our initial FEM simulations, only 4 designs were found with $CL < 0.335$ dB/km. The NN identified 362 designs (limited only by the time to run our FEM confirmations), allowing for statistically significant examination of the distributions in high-performance designs. The FEM high-performance

designs overlap the range of parameters in the NN results, but the volume of NN designs gives a wealth of diversity in parameters.

We will exploit the statistical relevance of the NN set in the next section to find a figure of merit for high-performance designs. The richness of the solution set can also be exploited for secondary optimizations, beyond our focus on CL and HOM suppression alone.

Figure of Merit for High-performance

We select $[g, D_{cap}, \gamma, D_{nest}]$ as our primary features heavily correlated with optimal performance. Other 4-parameter vectors were examined, but $[g, D_{cap}, \gamma, D_{nest}]$ was the most informative. In Fig. 4 we visualize the four dimensional designs, as mapped onto a 2D plane. This allows us to identify patterns, correlations, and clusters among different fiber design parameters.

Numerical comparison of the three datasets highlights the efficiency of the NN in identifying a specific high-performance region. The training set in Fig. 4a spans a vast parameter space with varied confinement loss. The reserved high-performance set in Fig. 4c has uniform color (all CL below 1 dB/km), yet it remains fairly dispersed. Contrastingly, the NN optimized designs in Fig. 4b exhibit a distinct "pinching" effect. This convergence suggests that $[g, D_{cap}, \gamma, D_{nest}]$ effectively define a narrow structural corridor essential for minimizing confinement loss. The NN successfully isolated this optimal configuration, whereas traditional sampling methods yielded more scattered results even within the low-loss regime.

We use a profile centroid FOM to quantify the statistical proximity of a design to the optimal performance regime. We start from our 362 NN-optimized designs. Each parameter in $[g, D_{cap}, \gamma, D_{nest}]$ is normalized using the mean

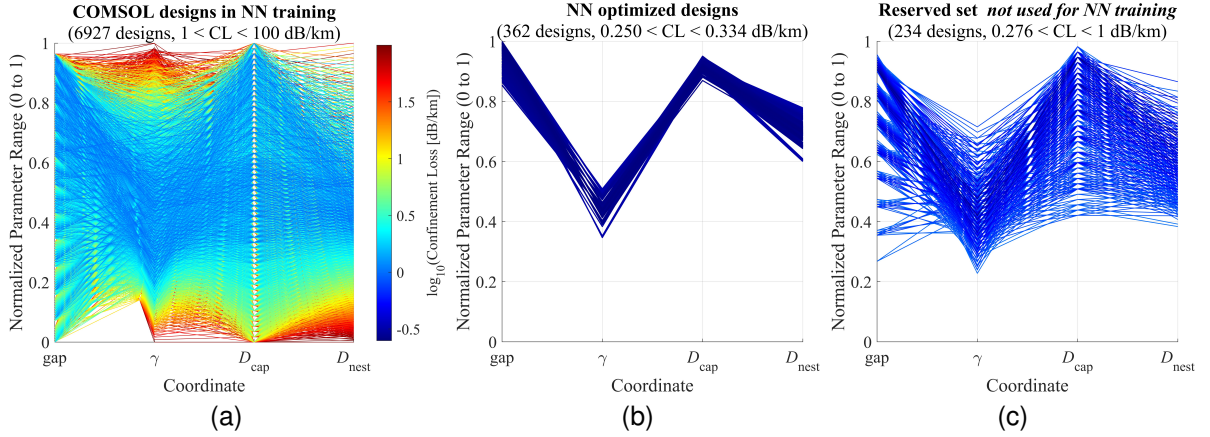


Fig. 4: Visualization of design performance in: (a) FEM dataset used for training, (b) neural network optimized designs, and (c) high-performance FEM designs reserved set not seen during training.

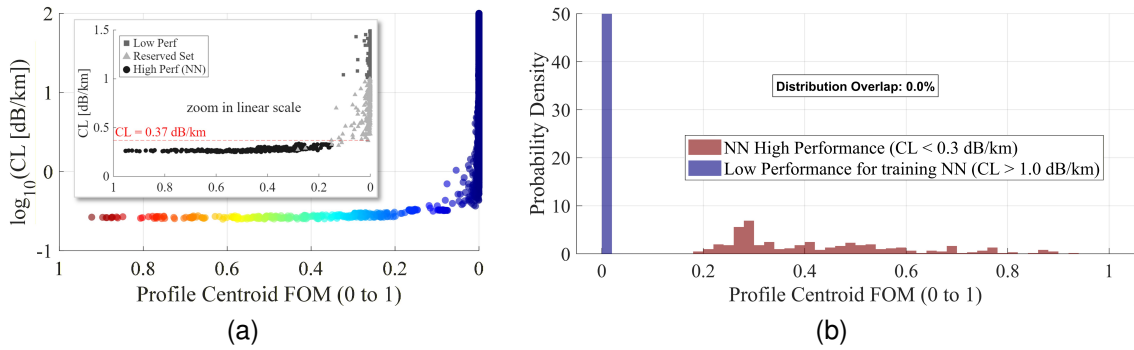


Fig. 5: a) Confinement loss vs. FOM (also in color map) for all designs, with a zoom in linear scale showing design set; and b) empirical FOM probability density for two sets.

(μ) and standard deviation (σ) within this set

$$\begin{aligned} z_{g,j} &= \frac{g_j - \mu_g}{\sigma_g}, & z_{D_{cap},j} &= \frac{D_{cap,j} - \mu_{D_{cap}}}{\sigma_{D_{cap}}}, \\ z_{\gamma,j} &= \frac{\gamma_j - \mu_\gamma}{\sigma_\gamma}, & z_{D_{nest},j} &= \frac{D_{nest,j} - \mu_{D_{nest}}}{\sigma_{D_{nest}}} \end{aligned} \quad (2)$$

where $\mu = [5.98, 51.9, 0.524, 24.3]$ with diameters and gap in μm , and $\sigma^2 = [0.109, 0.463, 0.0228, 1.05]$. The total deviation of design j from the centroid is the Euclidean norm, f_j , and we map that distance to a FOM

$$\begin{aligned} f_j &= \sqrt{z_{g,j}^2 + z_{\gamma,j}^2 + z_{D_{cap},j}^2 + z_{D_{nest},j}^2} \\ FOM_j &= \exp(-f_j^2/2). \end{aligned} \quad (3)$$

The FOM range is $[0, 1]$, where 1 represents the ideal centroid. In Fig. 5a we plot the FOM for all three sets of designs. The FOM is extremely peaked; Fig. 5b shows zero overlap between low and high performance designs.

Conclusion

Our machine learning framework effectively navigates the high-dimensional design space of NANF. The underlying NN provides the necessary computational efficiency, and this work brings out the statistical analysis of the resulting diverse dataset. The statistical depth of the NN results allowed us

to identify geometric hallmarks of low-loss regimes and develop a profile centroid FOM to quantify design proximity to the high-performance regime. Our results show a 0.0% distribution overlap in FOM between low-performance training data and the high-performance set, validating the predictive power of this data-driven analysis. This approach offers a scalable methodology essential for navigating complex future structures such as DNANF or multi-mode architectures.

References

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